



The Bureau
of Meteorology

Weather radar tutorial

Monash Radar Workshop
2025

BOM Radar science team





The Bureau
of Meteorology

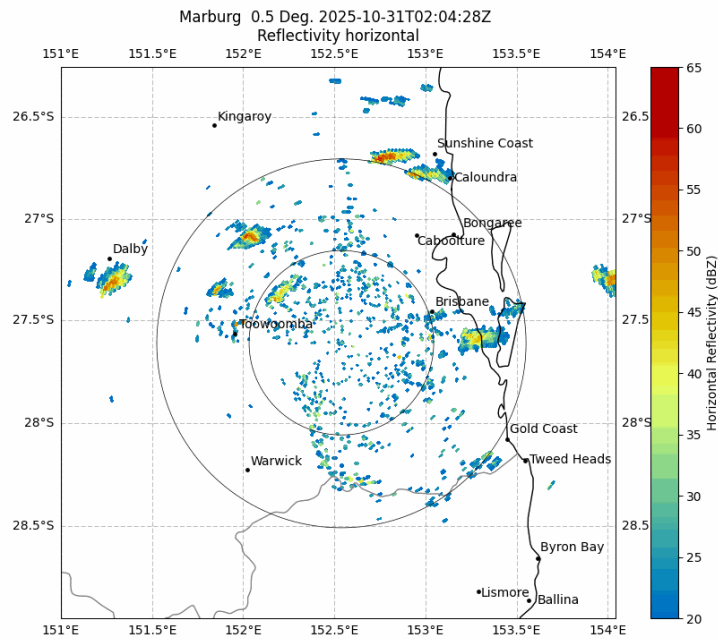
What are weather radars?

Examples of weather surveillance radars. Clockwise from upper left: The Denver (USA) area WSR-88D (KFTG) from outside; the antenna and pedestal of KFTG from inside the radome; a television-owned radar on top of the Rockefeller Center in New York City; a scanning radar system in the State of São Paulo, Brazil; and the S-Pol transportable research radar in Colorado.

(From Radar Meteorology by Fabry 2020).

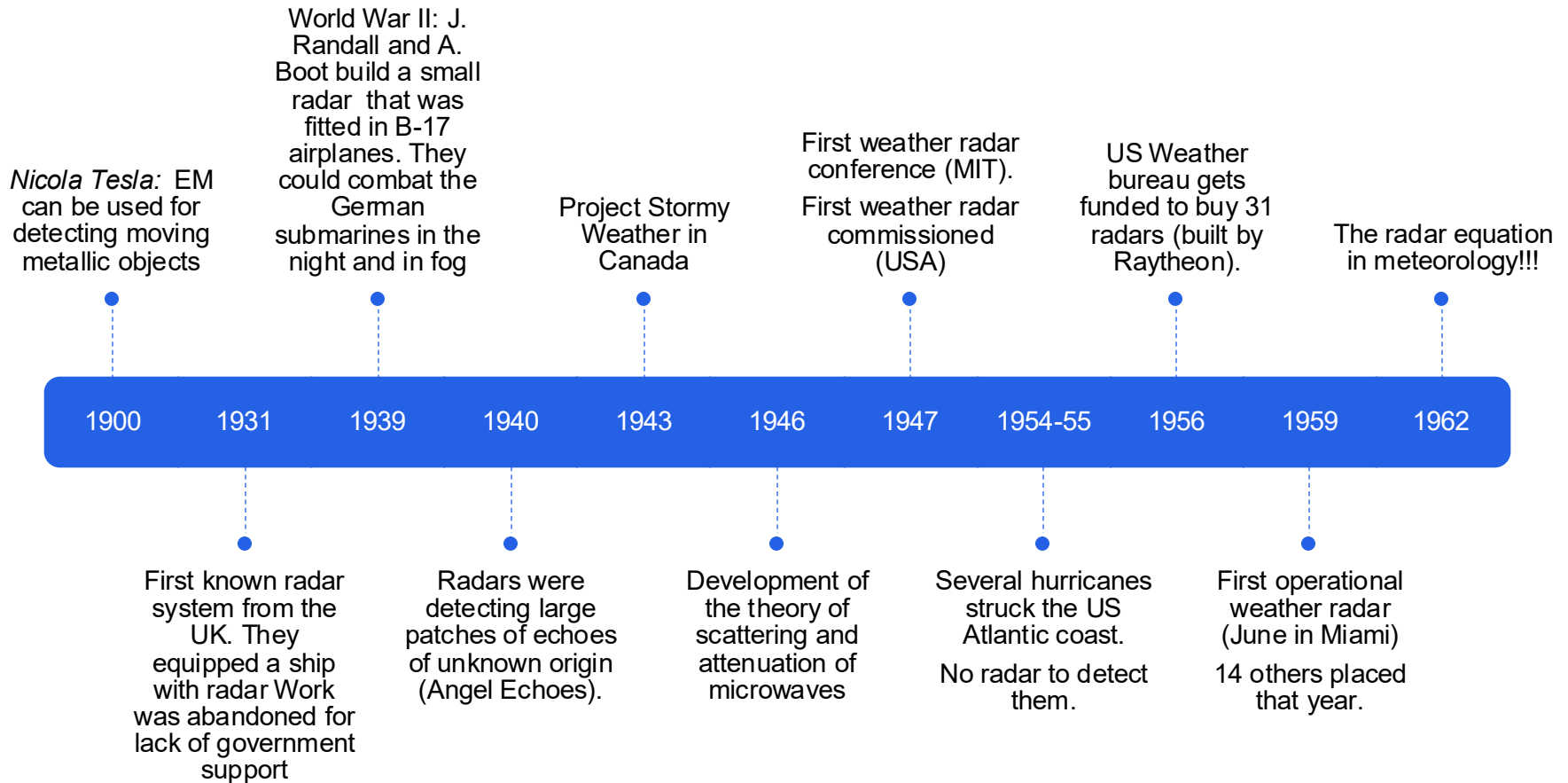


A dramatic weather radar loop



[Giant hail causes injuries and damage to cars, homes across south-east Queensland - ABC News](#)

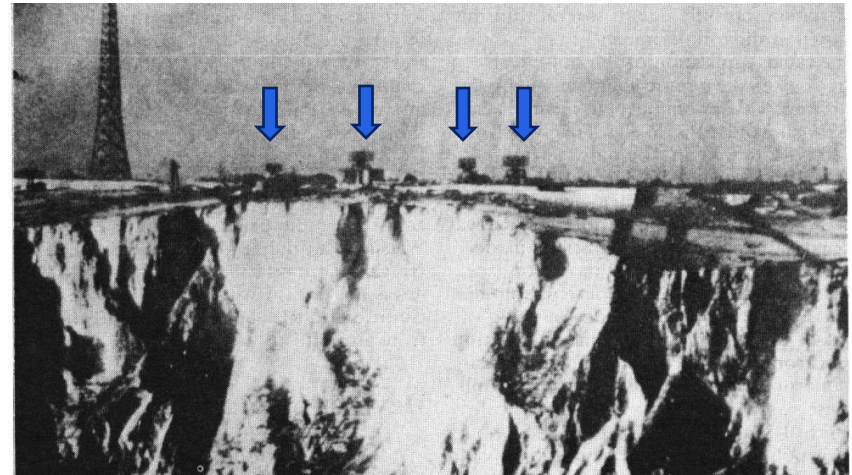
Short history of radars



Short history of radars

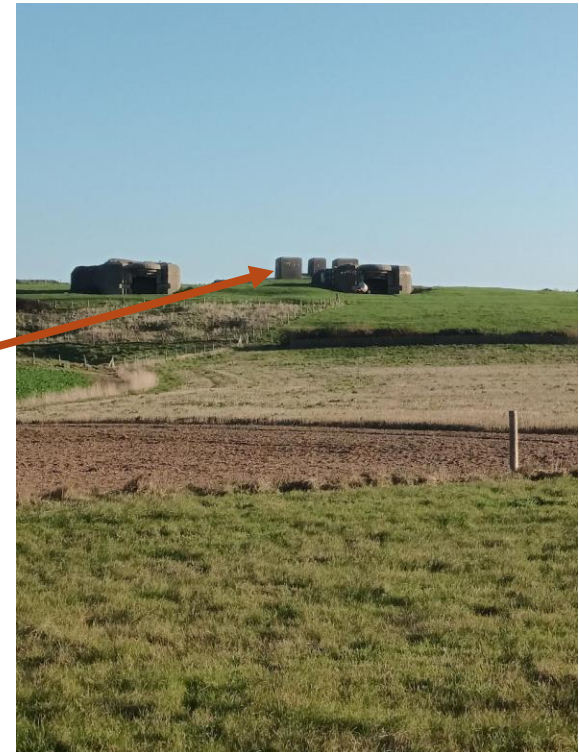
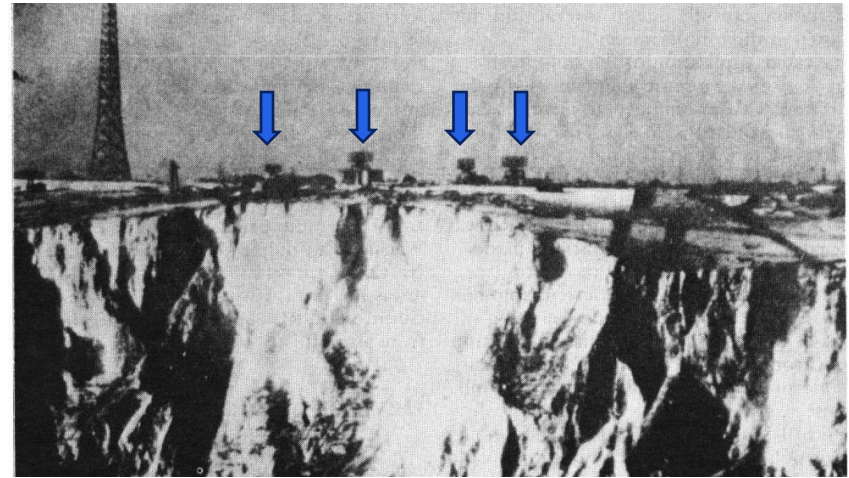


5 Men (US army) crew operating a SCR-268 radar during WW2

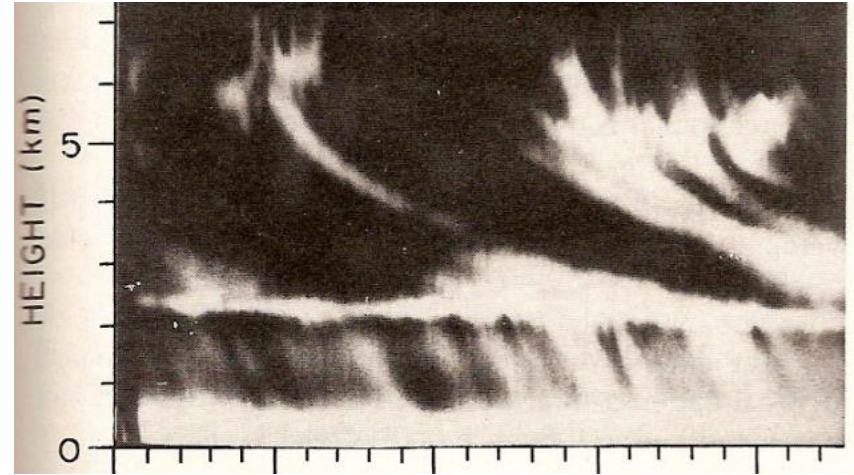


British coastline (near Dover) with radar towers. German photography with telephoto camera October 1940 from France side.

Short history of radars



Short history of radars



- The Stormy Weather Group (Canada)
- First Australian radar, operated by lighthouse staff.

Douglas, R.H. (1990). The Stormy Weather Group (Canada). In: Atlas, D. (eds) Radar in Meteorology. American Meteorological Society, Boston, MA. https://doi.org/10.1007/978-1-935704-15-7_8

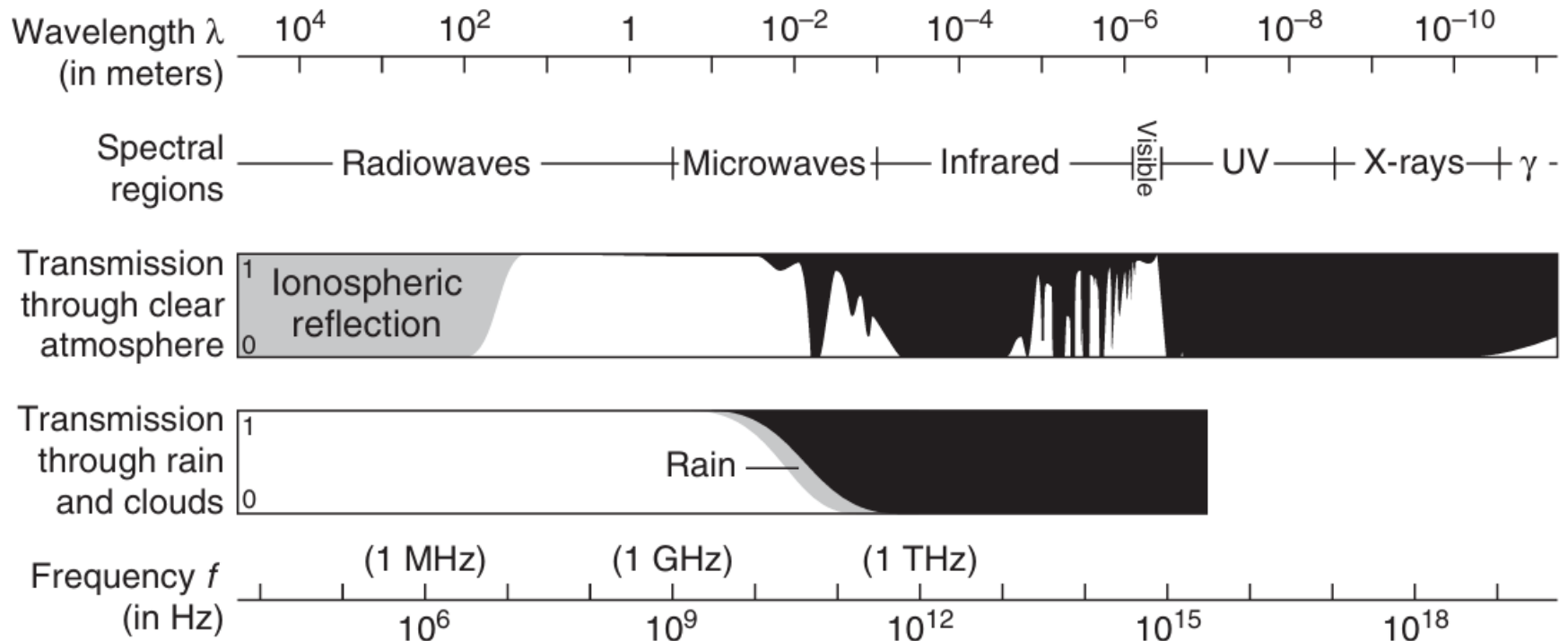




What radar measures?

What Radar Actually Measures?

Power of Return Signal → Reflectivity (Z)



Microwaves are a part of the electromagnetic spectrum where rain causes partial attenuation (**neither completely transparent nor completely opaque**) making them ideal for weather radar applications



What Radar Actually Measures?

Power of Return Signal → Reflectivity (Z)



- "How much energy came back?"
- Tells us: **How much stuff is out there? How big are the particles?**
- Units: dBZ (decibels of Z):

$$Z = \int N(D) \cdot D^6 dD$$

This is the distribution of particles times the 6th power of the particle size:

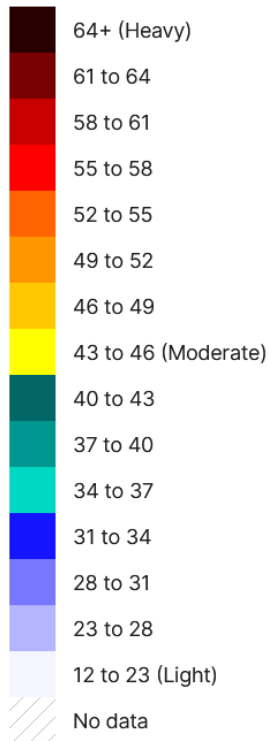
- **More particles = more signal.**
- **Bigger particles = a lot more signal.**



What Radar Actually Measures?

The scale of the reflectivity

Rain reflectivity (dBZ)

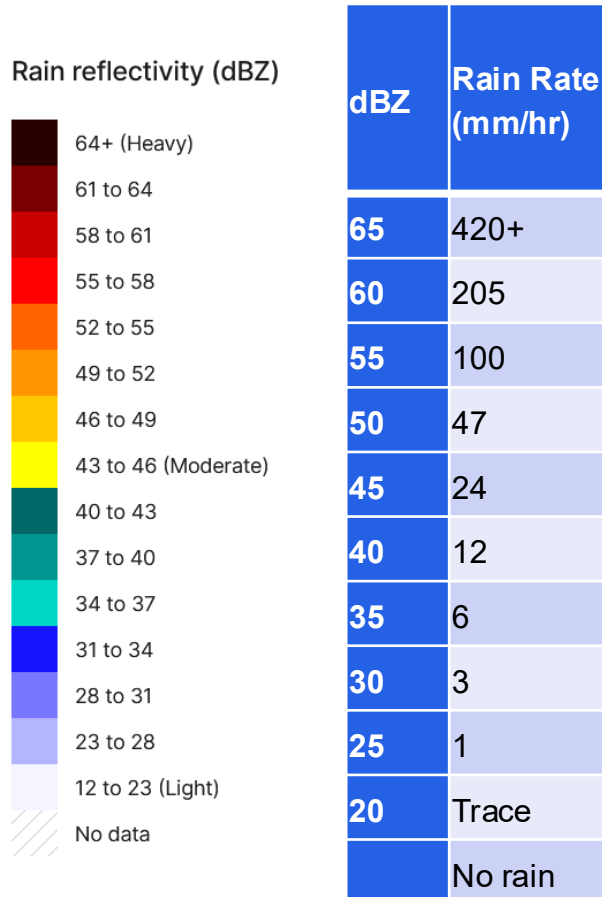


dBZ Range	Precipitation Type
-35 to 0 dBZ	Extremely light precipitation (drizzle or light snow)
0 to 20 dBZ	Very light precipitation or general clutter (light snow, very light rain)
20 to 40 dBZ	Light to moderate precipitation (light rain , moderate snow)
40 to 50 dBZ	Moderate to heavy precipitation (heavy rain , downpours)
50 to 65 dBZ	Heavy precipitation or some hail (heavy thunderstorms)
> 65 dBZ	Extremely heavy precipitation (intense thunderstorms with water-coated hail)



What Radar Actually Measures?

The dBZ values equate to approximate rainfall rates indicated below.



The Z–R relationship developed by J. S. Marshall and W. M. Palmer (1948) is traditionally used to transform reflectivity into rainfall rate:

$$Z = 200 \cdot R^{1.6}$$

Where:

$$Z_{dBZ} = 10 \cdot \log_{10} \left(\frac{Z}{1 \text{ mm}^6 \cdot \text{m}^{-3}} \right)$$

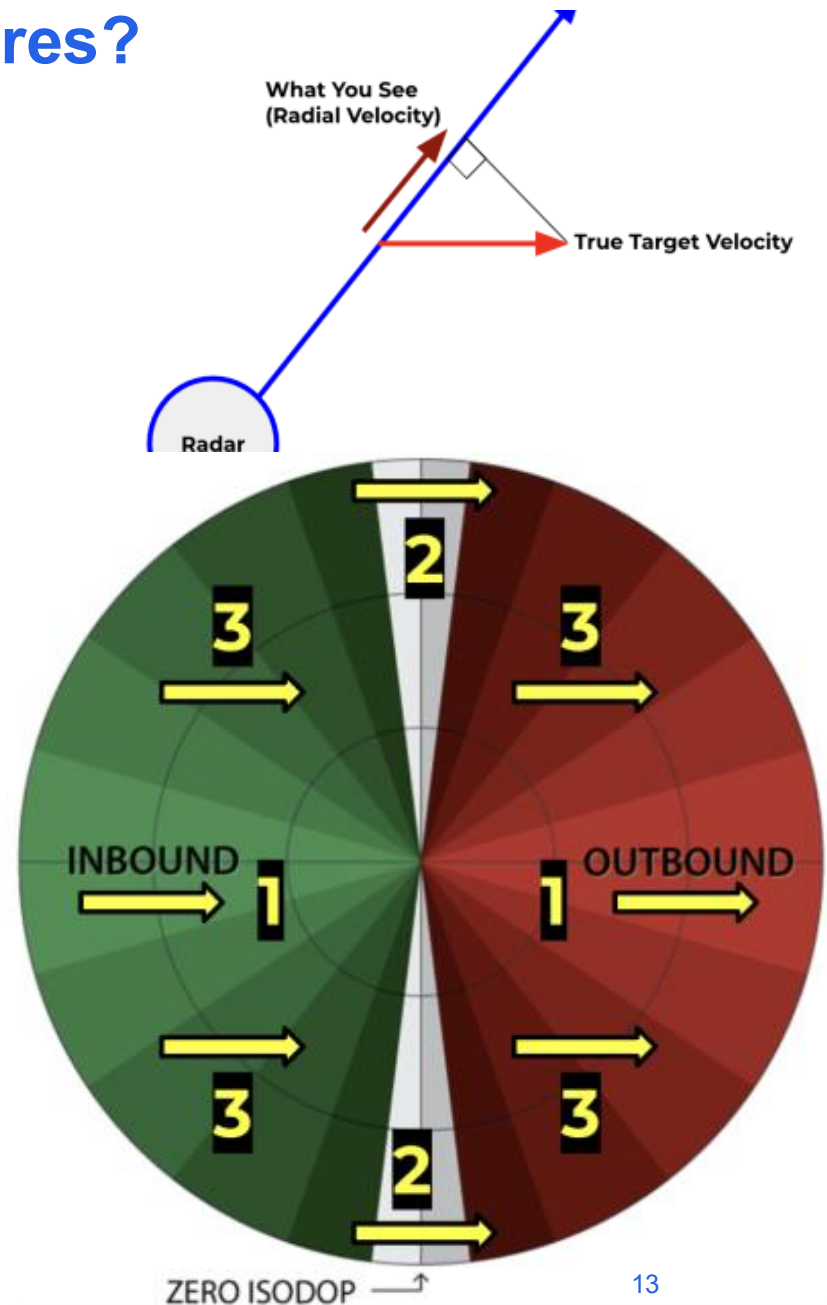


What Radar Actually Measures?

Doppler Velocity

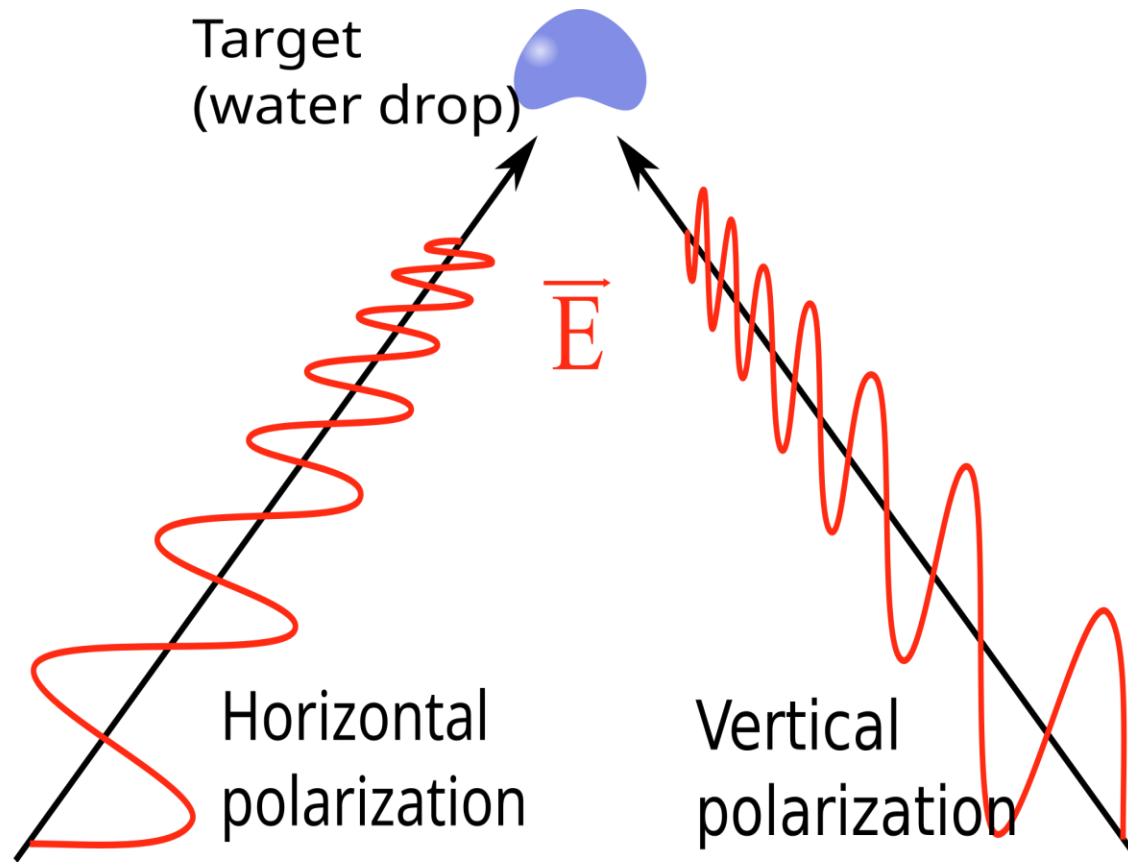
Velocity is the second of the three base products that are produced by pulsed Doppler radars and is used to indicate the motion and speed of targets. Since the **radar is at a fixed location**, it can only measure how fast a **target is moving toward or away from the radar itself**. This is known as radial velocity, and it differs from true velocity.

- Radial velocity will always be **less than or equal to** actual target velocity.
- Radial velocity will be equal to the actual target velocity **only** when the target motion is directly toward or away from the radar.
- When the target motion is **perpendicular to the radar beam**, the radial velocity will be **zero**.



What Radar Actually Measures?

Dual-Polarisation: What the shape and orientation of particles tell us



What Radar Actually Measures?

Dual-Polarisation: What the shape and orientation of particles tell us

Differential Reflectivity (ZDR)

Compares horizontal vs. vertical returns

Tells us:

Particle shape and size

- Large drops = high

ZDR (flat)

- Hail = low ZDR (round)

- Rain vs. ice discrimination

Correlation Coefficient (ρ_{HV})

Similarity of H and V pulse returns

Tells us:

Uniformity of particle types

- ~ 1.0 = pure rain

- $\sim 0.85-0.95$ = melting/mixed

- Low values = debris, birds

Specific Diff. Phase (KDP)

Phase shift rate through particles

Tells us:

Rainfall rate and content

- High KDP = heavy rain

- Unaffected by attenuation

- Identifies rain vs. hail cores

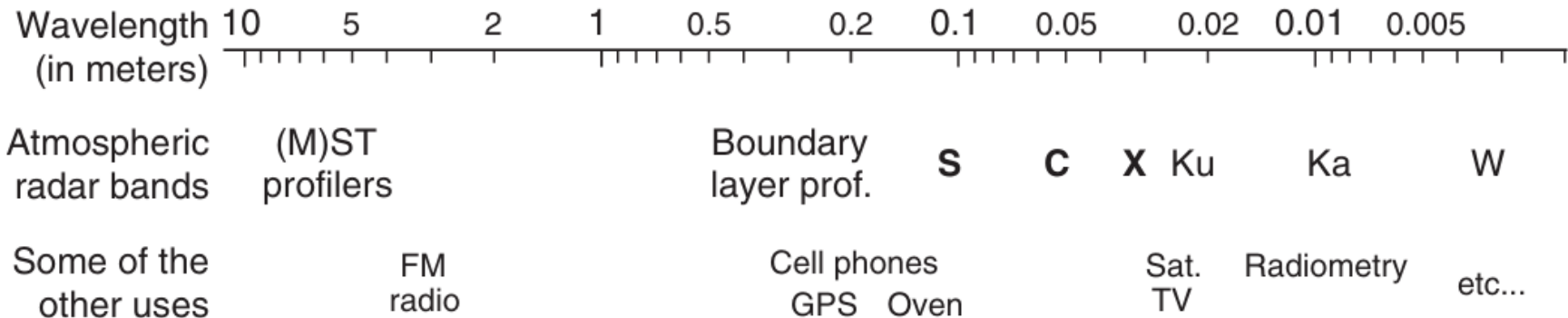
Key Insight: By sending radar pulses in both horizontal and vertical orientations, we can identify what type of precipitation is falling, not just how much





Types of Weather Radars

Types of Weather Radars



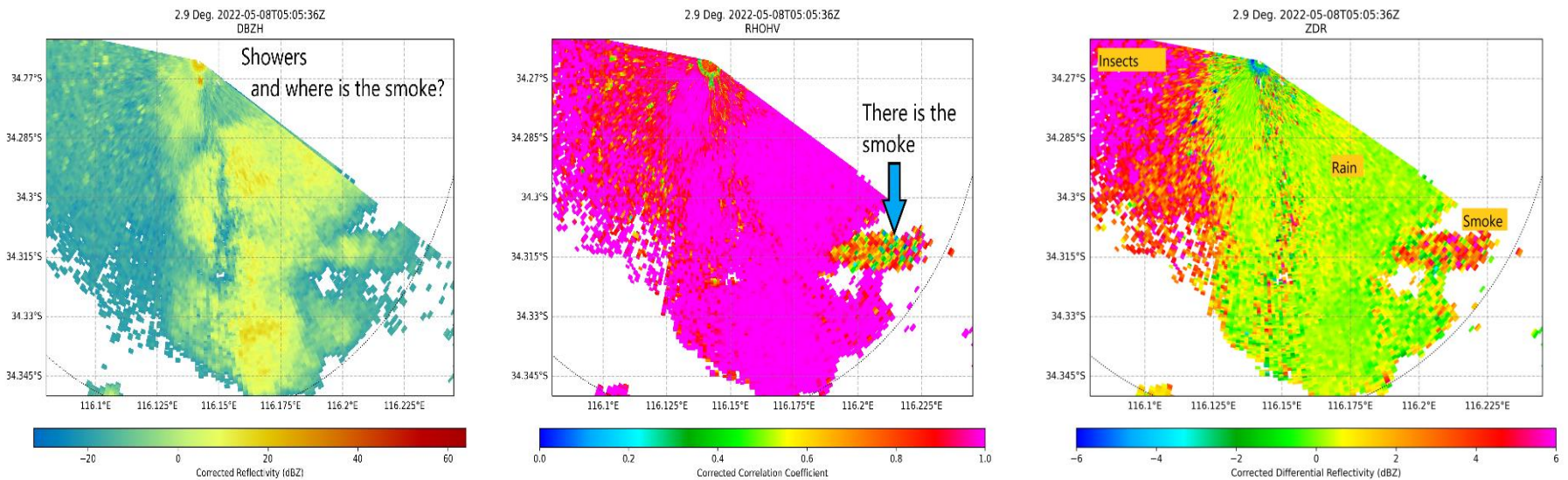
- Precipitation**
 - **S-band (~10 cm wavelength, ~3 GHz):** Long range, minimal attenuation, penetrates heavy rain well (NEXRAD/WSR-88D)
 - **C-band (~5 cm wavelength, ~5 GHz):** Balance of range and resolution, moderate attenuation (many international networks)
 - **X-band (~3 cm wavelength, ~10 GHz):** High resolution, portable, but significant attenuation in heavy rain (research, gap-filling, urban networks)
- Clouds**
 - **Ku/Ka bands (~2 cm/0.8 cm wavelength, ~15/35 GHz):** Very high resolution, severe attenuation, used for cloud microphysics and spaceborne radar (GPM)
 - **W band (~3 mm wavelength, ~94 GHz):** non precipitating and lightly precipitating cloud profiles



X-Band: Wildfire Monitoring



X-Band: Wildfire Monitoring



Guyot et al. (2023, AMT, submitted) : using dual-pol moments and their texture to separate clear air (insects), bushfire plume and hydrometeors (including pyroCu and pyroCb).

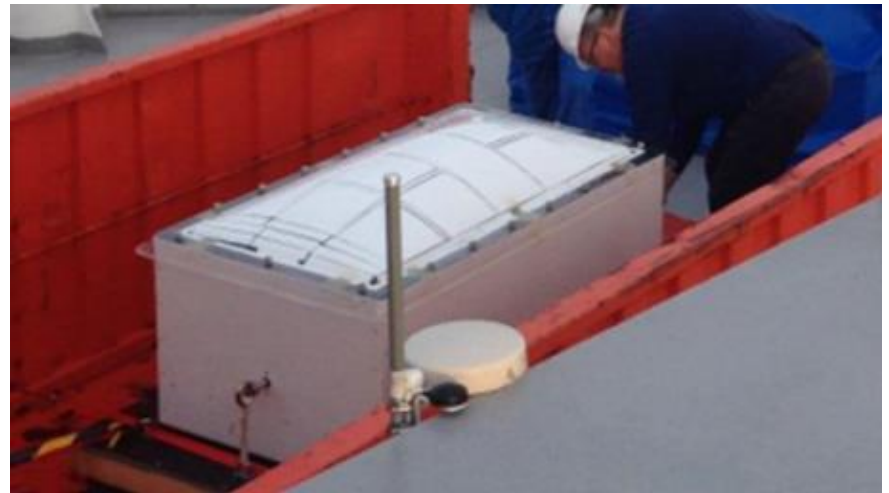
Ka/Ku band radars



W-band: cirrus clouds

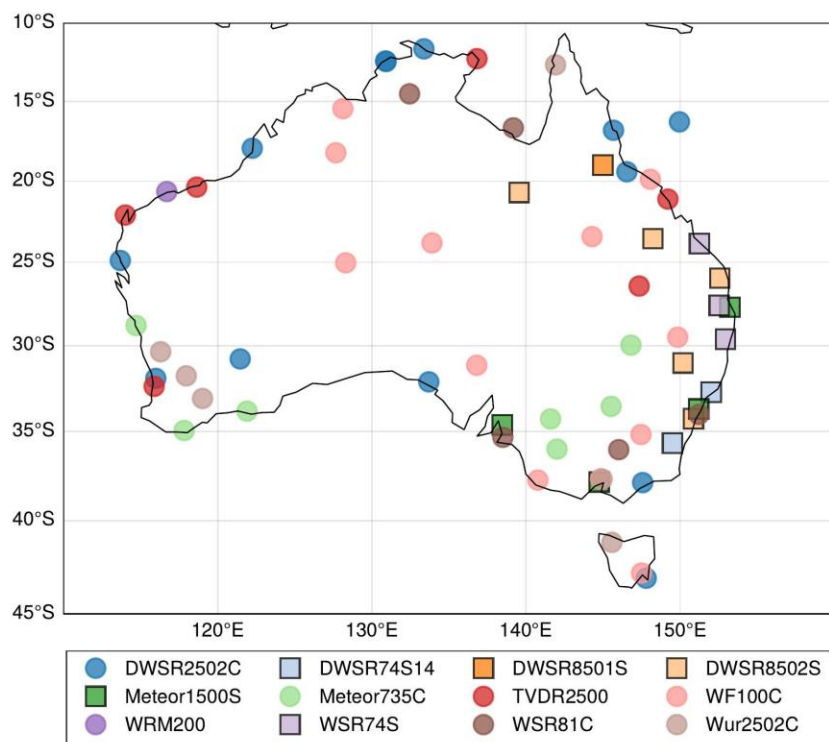
BASTA radar

Non precipitating and lightly precipitating
cloud profiles



State of the Australian weather radar network.

- The Australian weather radar network:
 - **70** (and growing) radars.
 - **20 different types** (Manufacturers, beamwidth, bands, PRF, dual-pol, ...)
 - S, C, and soon X-bands





An Overview of Py-ART

History of the Py-ART

- Development started at the BoM by Scott Colis.
- It began to address the needs of ARM with the acquisition of new scanning cloud and precipitation radars as part of the American Recovery Act.
- The project has since expanded to work with a variety of weather radars and a wider user base including radar researchers and climate modelers.
- The software has been released on GitHub as open source software under a BSD license. Runs on Linux, OS X. It also runs on Windows with more limited functionality.

What can PyART Do?

- Py-ART can be used for a variety of tasks from basic plotting to more complex processing pipelines. Specific uses for Py-ART include:
- Reading radar data in a variety of file formats.
- Creating plots and visualization of radar data.
- Correcting radar moments while in antenna coordinates, such as:
 - Doppler unfolding/de-aliasing.
 - Attenuation correction.
 - Phase processing using a Linear Programming method.
- Mapping data from one or multiple radars onto a Cartesian grid.
- Performing retrievals.
- Writing radial and Cartesian data to NetCDF files.



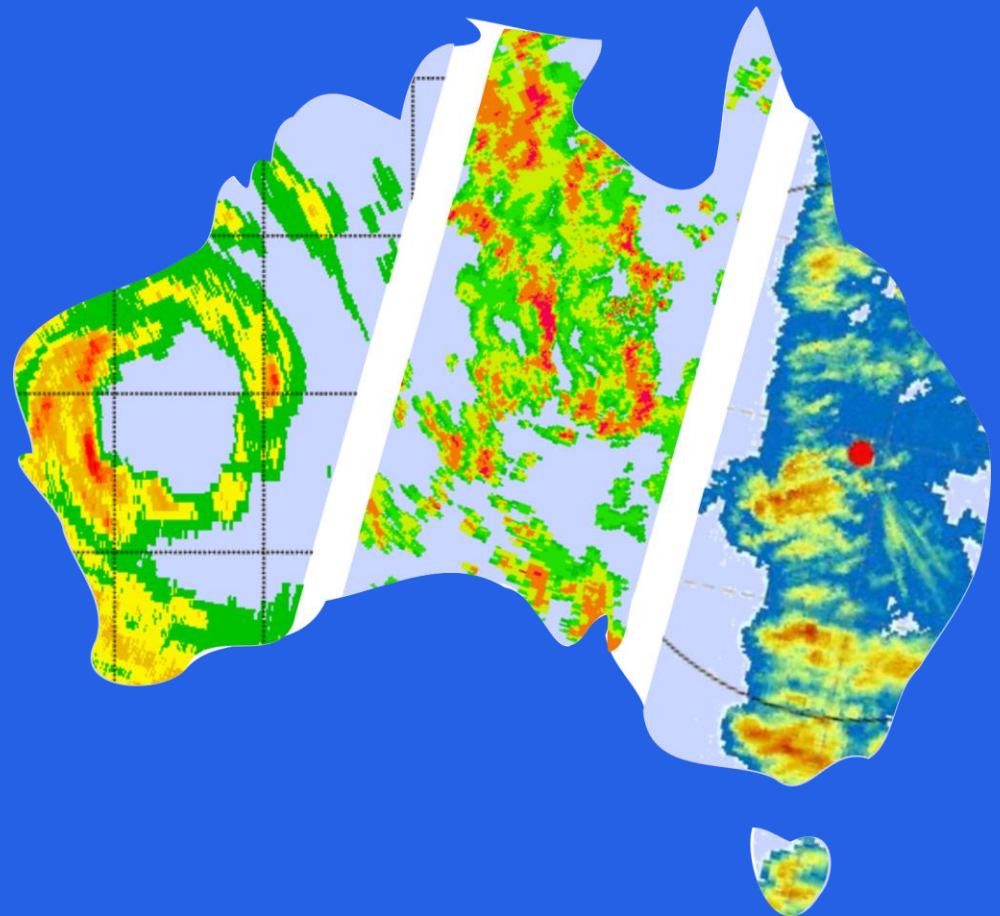


The Bureau
of Meteorology

Australian Unified Radar Archive

Monash Radar Workshop
2025

BOM Radar science team



What is AURA?

Radar data standardisation and accessibility

- Provides research and commercial users with public access to Australia's long-term operational weather radar archive and several research radars.
- Data converted into consistent open formats/specifics and product levels aimed at different users.
- Implements best practice approaches for QC, correction and gridding to reduce workload on users.
- Documentation (we are still trying to improve this one!)
- Connecting users of radar data
- Hosted on NCI under their National Reference Dataset program (300 TB) - simple and fast access for public and researchers



Motivations

 **Missed Opportunities:** Radar data is avoided because it's too difficult to access and apply

 **Consistency:** Published literature is difficult to compare because QC and correction procedures differ.

 **Flawed Analysis:** Critical QC and corrections are not implemented and therefore published results are possibly flawed

Datasets

Level 1: Uncalibrated & uncorrected data in a common data format (odimH5, netCDF4) and standardized naming

Level 1b: Calibrated & corrected data in both 3D Cartesian grids (netCDF4) and volumes (cfradial netCDF4)

Level 2: 2D Cartesian grids for radar fields and retrievals at specific altitudes above radar level (netCDF4). Possible products include: echo top heights, maximum columnwise reflectivity, classification, rainfall, azimuthal shear

QPE: Rainfall Products generated either by the operational Rainfields3 system or research pipelines using Rainfields3 functionality (ground reflectivity, rain rate, rain accumulation as netCDF4)

Level 1b and level 2 QC description: Bowden et al. (2024). Identification of rainfall events and heavy. Identification of rainfall events and heavy rainfall events from radar measurements in southeastern Australia. JGR



Datasets

Level 1: Daily zip of odimh5 or cfradial for each radar

Level 1b: Daily zip of netCDF4 3D grids or cfradial for each radar

Level 2: Daily netCDF4 files for each radar product (no zip!)

QPE: Daily zip of netCDF4 2D grids for each radar + national domains (310)



Collections – Research Radars



Monash X-band



CPOL



CP2



OceanPOL



UQXPOL



Collections – Research Radars

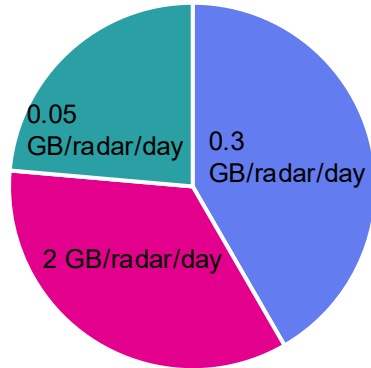
Published Datasets

- **OceanPol (RV Investigator, 2017 - present)**
 - Level 1b
- **CP2 (Brisbane, 2007 - 2015)**
 - Level 1
 - Level 1b
- **CPOL (Darwin, 1998 - 2017)**
 - Level 1b
 - Level 2
- **Monash Mobile Research Radar (WA project, Mobile)**
 - Level 1
- **UQXPOL (several projects, Mobile)**
 - Level 0 (manufacturer format)
 - Level 1

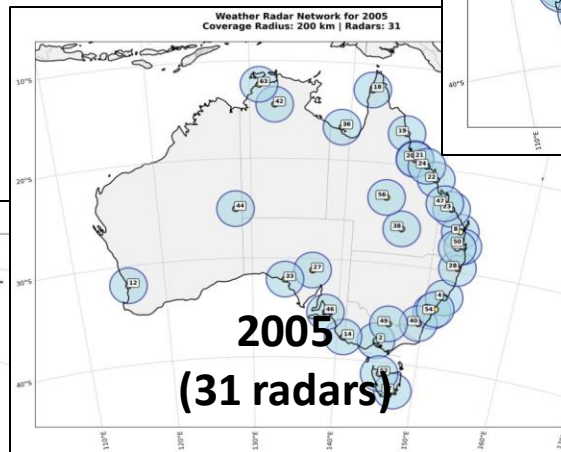
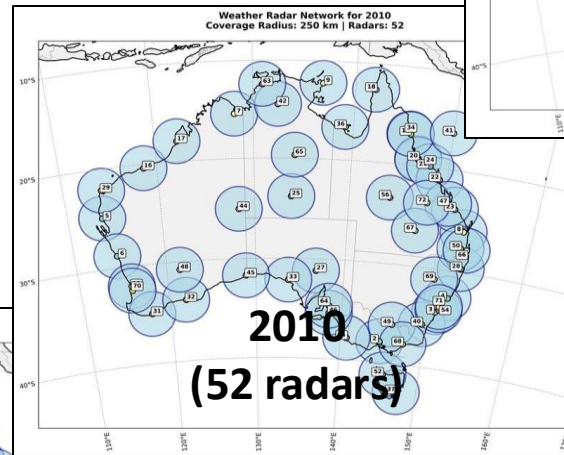
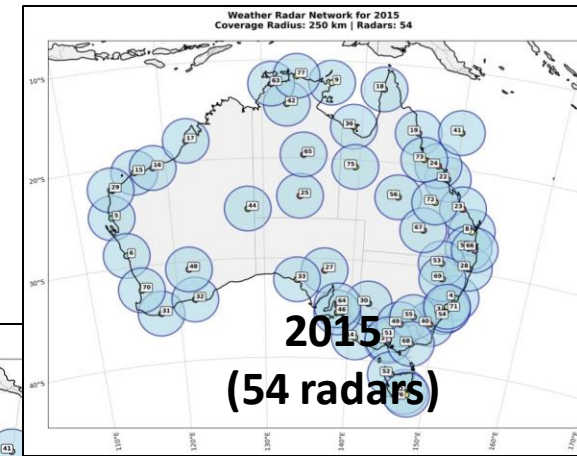


Collections – Operational Radars

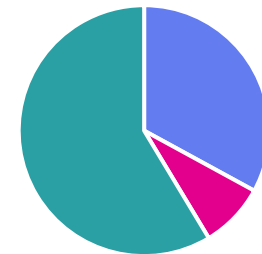
Level 1 active radar types (72)



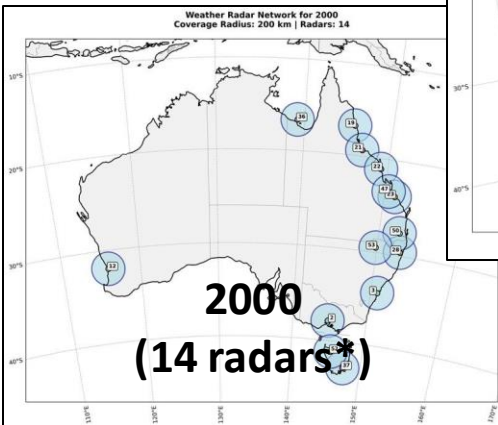
■ Doppler (30) ■ DualPol (25) ■ SinglePol (17)



Level 1 archive length (1567 radar years)



■ Doppler ■ DualPol ■ SinglePol



*with near complete records in AURA

Collections – Operational Radars

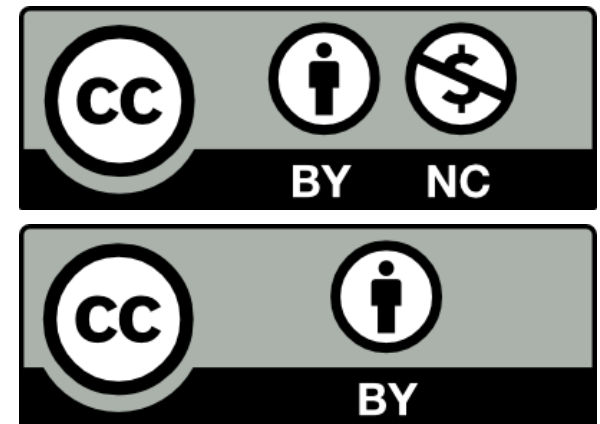
- **Published Level datasets**
 - Level 1 (updated daily from operational server)
 - Level 1b (updated when ERA5 data is ready)
 - Level 2 (updated when ERA5 data is ready)
- **Published QPE datasets**
 - Historical ground reflectivity (1998-2022)
 - Rainfields 3 (2019-present, updated daily from operational server)
 - Ground reflectivity
 - 15 min accumulation
 - Rain rate



Data Use Licensing and Attribution

- CC4.0-BY (Attribution is required)
 - Operational network level 1 dataset
- CC4.0-BY-NC (Attribution + non-commercial use)
 - All other datasets

NCI assigned doi provided for each dataset



NCI OPUS Portal

Radar Products

Created by  Jo Croucher, last updated by Joshua Soderholm on Oct 13, 2025 • 2 minute read

Overview

This collection contains a long-term archive of the Bureau of Meteorology operational weather network data. These observations are collected from sites across Australia for a period of more than twenty years and provide an invaluable understanding of precipitation and severe weather, including tropical cyclones and thunderstorms. **Level 1** data is unprocessed, ungridded (spherical coordinates) and provided using the Eumetnet **ODIMH5 model** of the HDF5 format. **Level 1b** consists of corrected and calibrated 3D gridded data and volumetric in netCDF4 format. **Level 2** data provides 2D grids of retrievals in netCDF4.

Summary

- **Operational:** Only weather radar data from [national network](#) is hosted
- **Updated:** Level 1 is updated regularly. Level 1b and 2 are updated infrequently.
- **Licensed:** Level 1 is licenced under [CC BY 4.0](#). Level 1b and 2 are licenced under [CC BY-NC 4.0](#). See licence document in the root directory of each dataset.

Citing

- **Level 1:** Soderholm J., A. Protat, C. Jakob, 2019: Australian Operational Weather Radar Level 1 Dataset. electronic dataset, National Computing Infrastructure, doi: [10.25914/508X-9A12](https://doi.org/10.25914/508X-9A12)
- **Level 1b:** Soderholm J., V. Louf, J. Brook, A. Protat, 2022: Australian Operational Weather Radar Level 1b Dataset. electronic dataset, National Computing Infrastructure, doi: [10.25914/40KE-NM05](https://doi.org/10.25914/40KE-NM05)
- **Level 2:** Soderholm J., V. Louf, J. Brook, A. Protat, R. Warren, 2022: Australian Operational Weather Radar Level 2 Dataset. electronic dataset, National Computing Infrastructure, doi: [10.25914/JJWZ-0F13](https://doi.org/10.25914/JJWZ-0F13)

Example Data Availability Statement: SHI data is available from the AURA level 2 dataset on NCI under a [CC BY-NC 4.0](#) licence from <https://dx.doi.org/10.25914/JJWZ-0F13>

This project was undertaken with the assistance of resources and services from Monash University, National Environmental Science Program, the National Computational Infrastructure (NCI) and the Bureau of Meteorology.

© Bureau of Meteorology 2025



Bureau of Meteorology weather radar network as of June 2025
(Bruce Gunn 2025)

Resources

- [Level 1](#) NCI Data Catalogue
- [Level 1b](#) NCI Data Catalogue
- [Level 2](#) NCI Data Catalogue
- [Example Python Code](#): How to access and analyse level 1 data

Get help

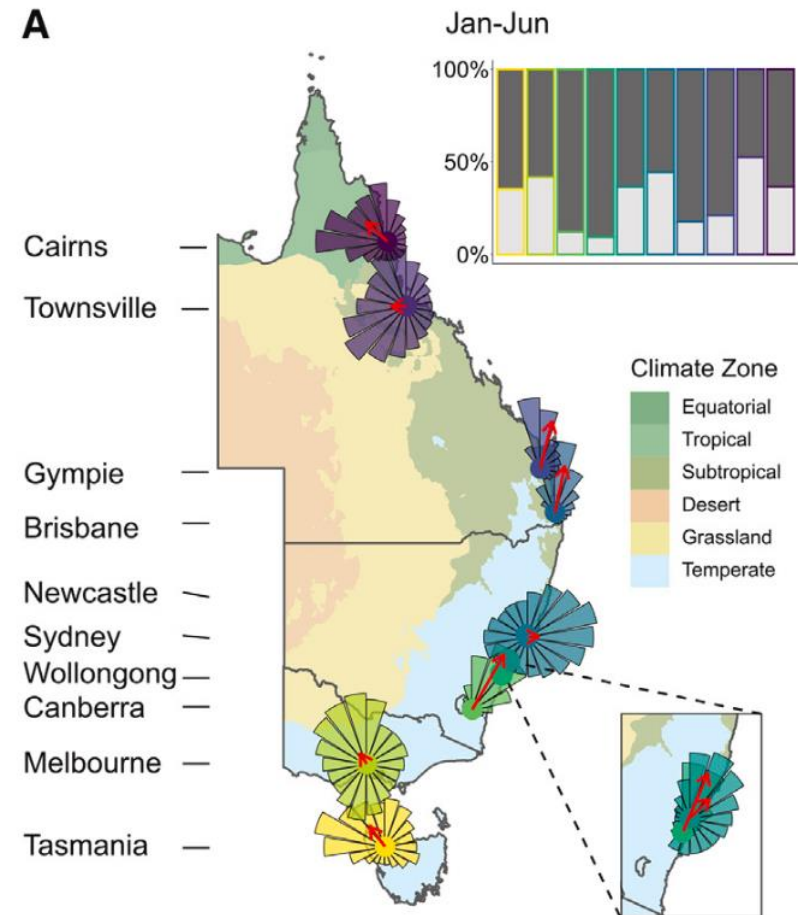
- View the parent [NCI Data Catalogue](#) entry
- Find the [identification number](#) of a radar (rq0 project)
- Questions? Email aurahelp@bom.gov.au

Openradar.io currently redirects to OPUS, but this web address will not be in use from in early 2026.



Applications (some highlights)

- Long-term changes in thunderstorm hazard frequency/intensity (e.g., Raupach et al. 2023, Hooman et al. 2022)
- Development of new hail retrievals to estimate damage (Ackermann et al. 2024)
- ML training and evaluation (ongoing across multiple projects)
- High resolution soil moisture (Weir 2024)
- Bushfire plumes and precipitation regimes (e.g., McGowan et al. 2025, Bowden et al. 2025)
- Ecological tracking and evaluation (bats, birds + insects, Shi et al. 2024)
- Space applications (meteorites and satellite/rocket reentry)
- Public health (thunderstorm asthma)



Migratory bird flight directions (Shi et al. 2024)

Future

Addition of historical and operational BRAIN datasets
(national blended gauge, satellite and radar QPE products)

Addition of SWIRL/3D winds products (historical + realtime to
assess wind hazard)

Automate production of operational level 1b and level 2
products (as ERA5 data is released)

Suggestions or requests? Contact us!





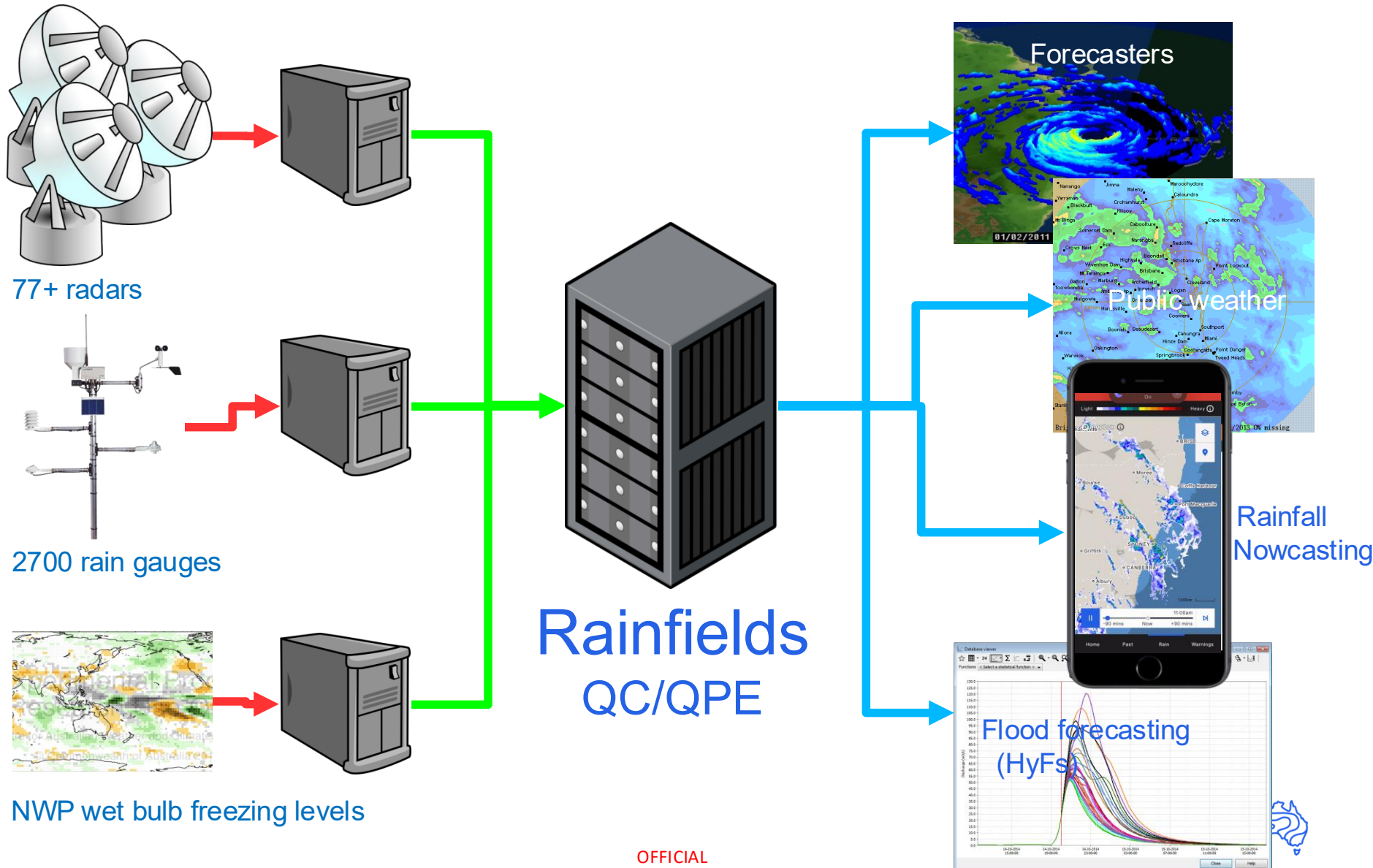
The Bureau
of Meteorology

Rainfields

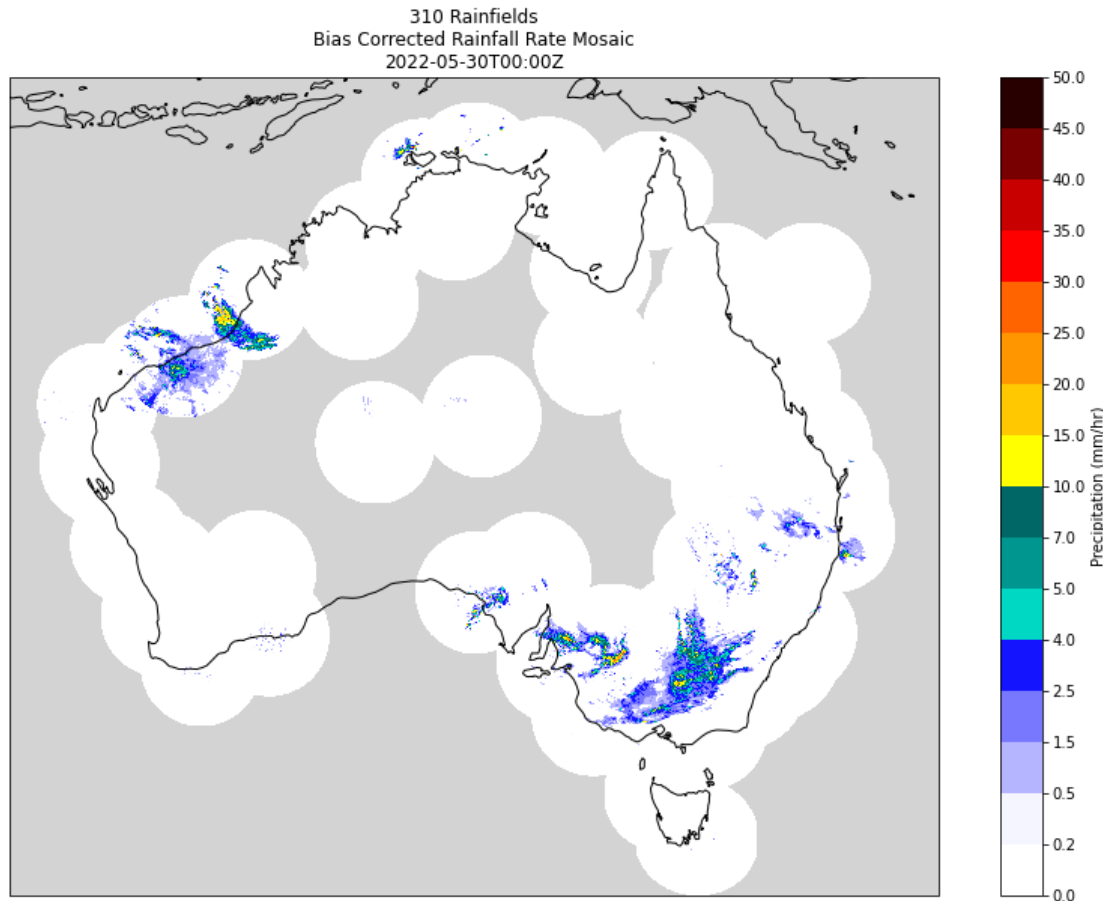
Monash Radar Workshop
2025

BOM Radar science team

RAINFIELDS : Real-time Rainfall Analysis



RAINFIELDS : Real-time Rainfall Analysis



- Radar Rainfall (every 5-min)
- Quality controlled radar volume scans (QC added to the volume scan data)
- Surface radar reflectivity for single and multi-radar domains
- Calibrated Instantaneous rain rate (used for accumulations and nowcasts)
- Rainfall accumulations
- Radar + gauge blends every 15 minutes (less extent)
- **High spatial and temporal resolutions**
- *Limited Coverage*

RAINFIELDS : Real-time Rainfall Analysis

Rainfields creates the following products for each radar in the network:

Quality Control and Non-Precipitation Related

- Cleaned Reflectivity
- Dealiased Doppler Velocity
- Hydrometeor Classification
- Vertical Profiles
- Equivalent Reflectivity at Surface
- Advection Vector Field
- Hail Diagnostics

Quantitative Precipitation Estimation (QPE)

- Instantaneous Rain Rate
- Gauge Calibrated QPE
- Gauge Merged QPE
- QPE as Average Recurrence Intervals
- Catchment Averages
- QPE Verification and Monitoring

Rainfields creates multi-radar mosaics combining the same type of products for each radar.

Products are generated using the datasets available at the time of processing and are not updated if new data becomes available later.

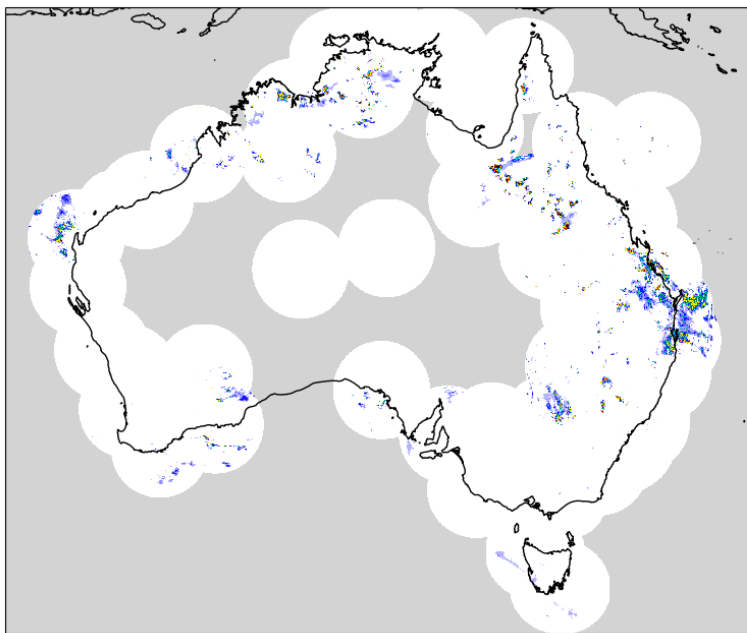


Products in AURA

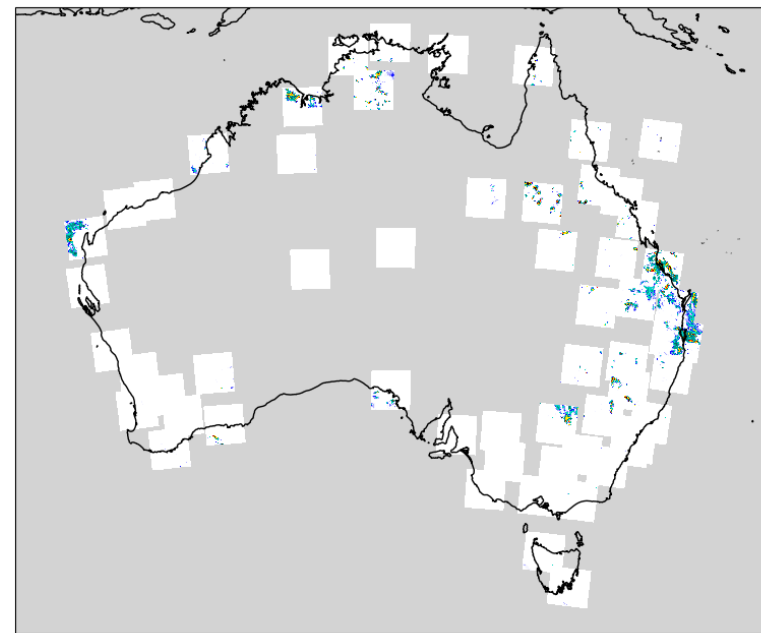
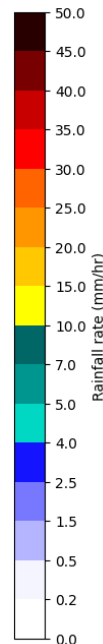
Rainfields Products - NCI Data Collections & Publishing - Opus - NCI Confluence

Per Radar - IDs (1, ..., 114) and **National multi-radar mosaic – ID 310**

- Equivalent Reflectivity at Surface (gndrefl)
- Gauge Calibrated rain rates (prcp-crate)
- Gauge Merged QPE (prcp-m15) *[limited extent]*



gndrefl and prcp-crate typical extent



prcp-m15 typical extent

